

- `git clone https://github.com/tensorflow/models`
- `cd models/research`
- `protoc object_detection/protos/*.proto --python_out=.`

The codes above will download the API. The next step should be to download all binaries from the system by the following code:-

- `mkdir -p /opt/graph_def`
- `cd /tmp`
- `for model in \`
 - `ssd_mobilenet_v1_coco_11_06_2017 \`
 - `ssd_inception_v2_coco_11_06_2017 \`
 - `rfcn_resnet101_coco_11_06_2017 \`
 - `faster_rcnn_resnet101_coco_11_06_2017 \`
 - `faster_rcnn_inception_resnet_v2_atrous_coco_11_06_2017`
- `do \`
 - `curl -OL http://download.tensorflow.org/models/object_detection/$model.tar.gz`
 - `tar -xzf $model.tar.gz $model/frozen_inference_graph.pb`
 - `cp -a $model /opt/graph_def/`
- `done`

Collect a good number of images that pertain to the object or objects that you want the system to identify. Get a ton of images online and have them saved preferably in the same folder as TensorFlow.

DATASET PREPARATION

On getting all the images, use an online annotation tool or just Adobe to pinpoint and annotate the object in the images that you want to be identi-

Model name	Speed	COCO mAP	Outputs
ssd_mobilenet_v1_coco	fast	21	Boxes
ssd_inception_v2_coco	fast	24	Boxes
rfcn_resnet101_coco	medium	30	Boxes
faster_rcnn_resnet101_coco	medium	32	Boxes
faster_rcnn_inception_resnet_v2_atrous_coco	slow	37	Boxes

COCO Datasets with Models From Google Developers